# Why model?

UCLA SOCIOL 114
Winter 2025
soc114.github.io

11 Feb 2025

# Arc of the course

- Working with data
- Inference without models
- Inference with models

# Learning goals for today (Part 1)

At the end of class, you will be able to

- explain the curse of dimensionality
- recognize the possible futility of nonparametric estimation

# Motivating a research question[1]

Income inequality across households depends on

1. inequality across individuals
2. how individuals pool into households

A college degree affects (1) and (2)

[1] Mare 1991, Schwartz 2013

# Research question

To what degree does finishing college increase the probability of having a spouse who finished college?

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**Data.** National Longitudinal Survey of Youth 1997

- Probability sample of U.S. non-institutional civilian youth age 12–16 on Dec 31 1996
- Surveyed annually 1997–2011, then biennially
- $n = 8{,}984$

# Data access

To access these data, first

- set your working directory where you will be working
- download two supporting files from us
  1. `nlsy97.NLSY97` is a tagset file containing the variable names
  2. `prepare_nlsy97.R` is an R script to prepare the data

# Data access

Now go to the data distributor

1. Register with the survey
2. Log in to the NLS Investigator
3. Choose the NLSY97 study
4. Upload the tagset `nlsy97.NLSY97` that you downloaded from us
5. In the Investigator, download the data. Type to change the file name from `default` to `nlsy97`
6. Unzip the file. Drag `nlsy97.dat` into the folder you will work in
7. In your R console, run the line of code below
   - this will take about 30 seconds to run
   - you will need these R packages: `tidyverse` and `Amelia`

```
source("prepare_nlsy97.R")
```

In the future, you can now load the data with

```
d <- readRDS("d.RDS")
```

# Register with the survey

## NLS Investigator

**Tell us about yourself** - Only email is required

First name:

Last name:

Organization:

Email: *

Confirm Email: *

**Enter your username and password** - All fields are required

Username: *

Username is automatically filled in from email field.

Password: *

Confirm password: *

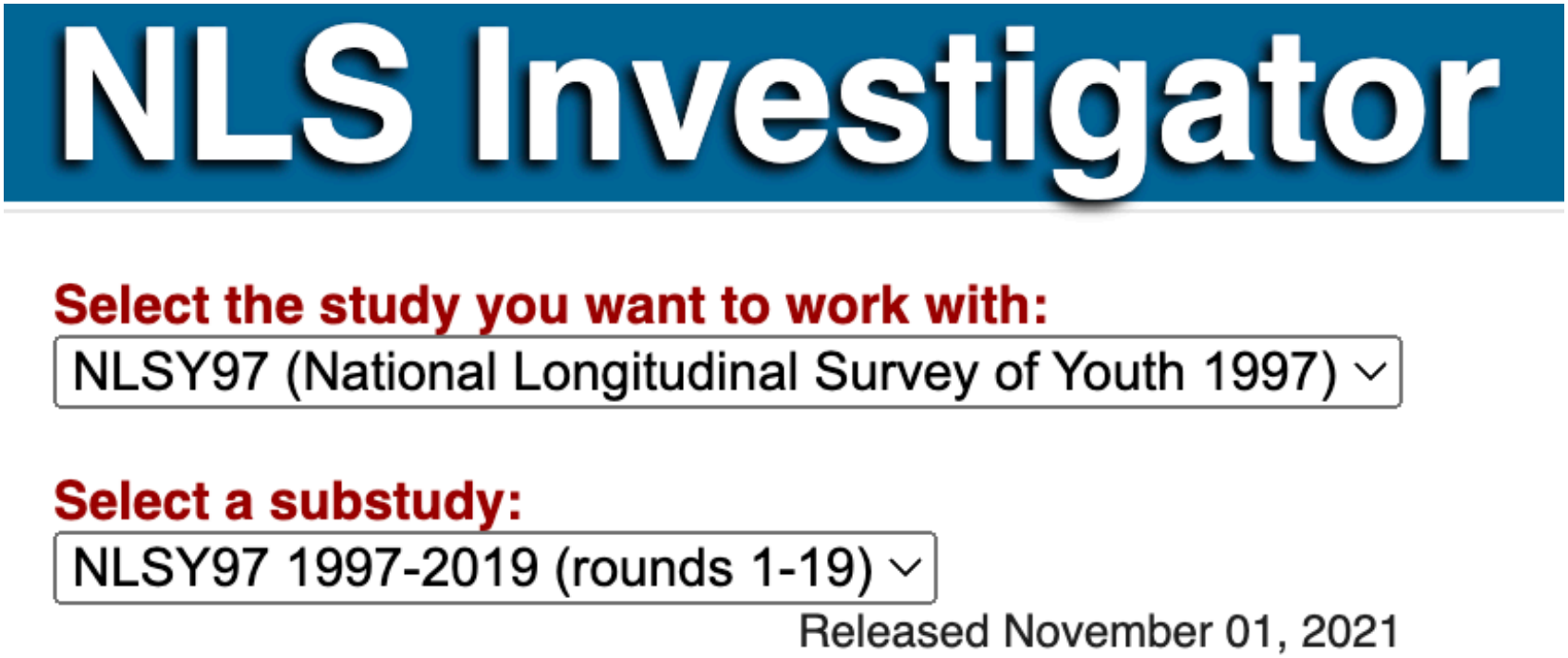
Password must be 8 characters or more and contain at least one numeric and one non numeric character.
In addition the password must not be based on username.

☐ I agree to the NLS Investigator Privacy Policy.

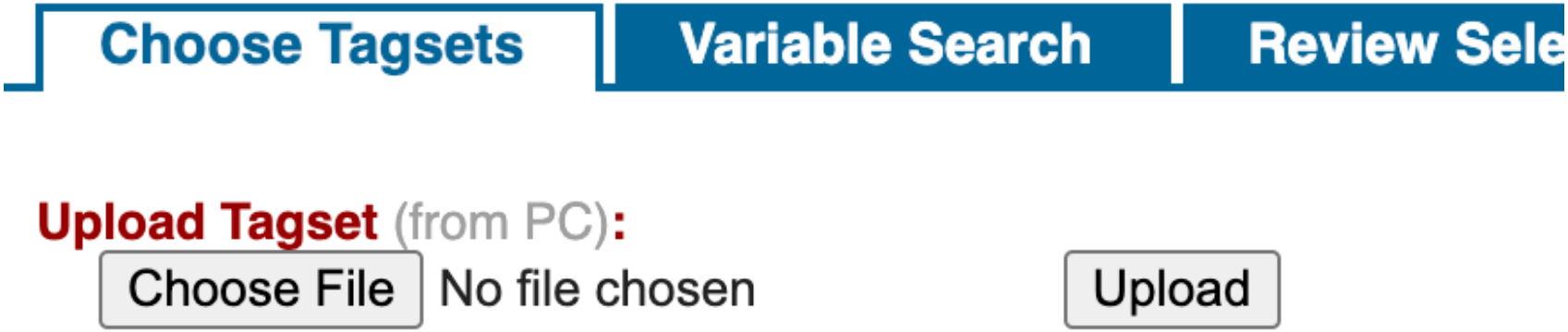
* Required field

Register

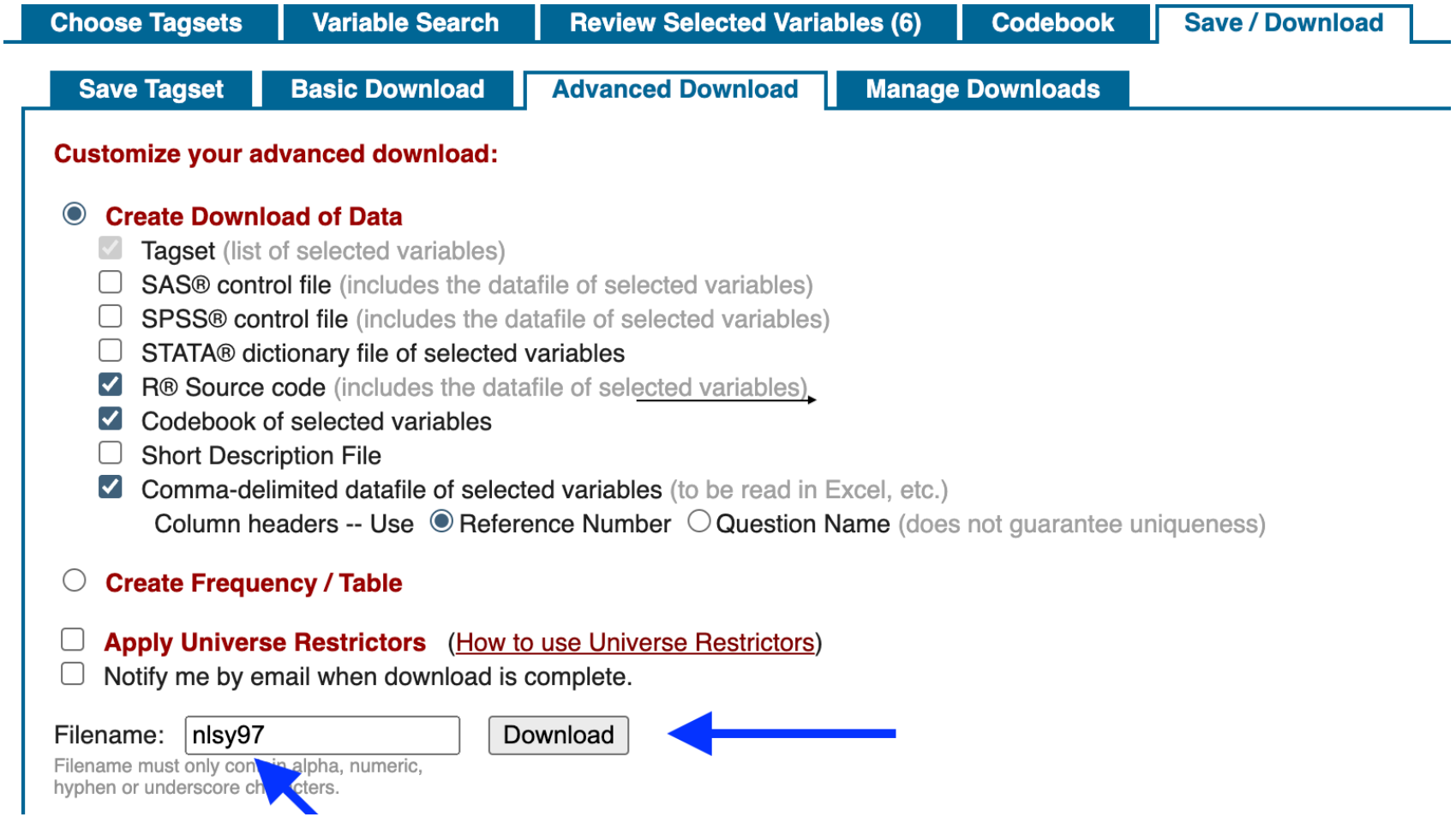
# Choose the NLSY97 study



# Upload our tagset



# Download the data



# Run our code

This code prepares the data file (one time, takes about 30 seconds)

```
source("prepare_NLSY97.R")
```

This code loads the prepared data (after the above, very fast)

```
d <- readRDS("d.RDS")
```

# Research question

To what degree does finishing college increase the probability of having a spouse who finished college?

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To what degree does finishing college increase the probability of having a spouse who finished college?

- Treatment $A$: Finished BA by age 25
- Outcome $Y$: Spouse or partner at age 30–40 holds a BA
  - 0 if no spouse or partner, or partner with no BA
  - 1 if spouse or partner holds a BA

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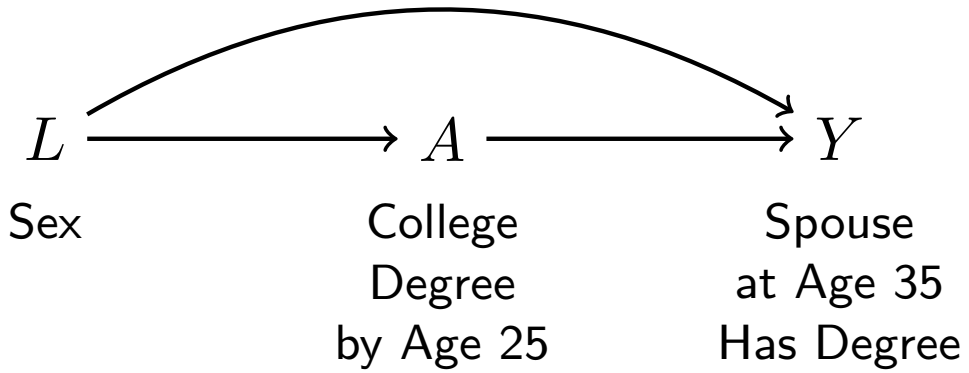

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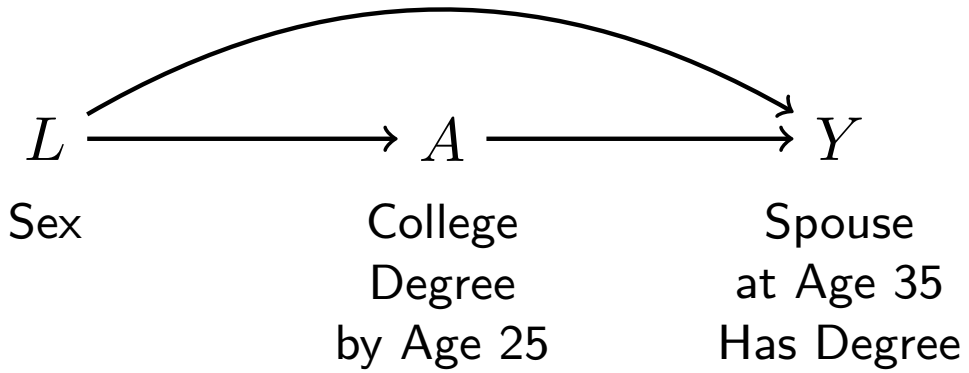


Adjustment procedure

1) Estimate within subgroups defined by {sex}
2) Aggregate over the subgroups

# Data

```r
d %>%
  select(sex, a, y) %>%
  print(n = 8)
```

```
# A tibble: 7,771 x 3
  sex    a          y
  <chr>  <chr>      <lgl>
1 Female college    FALSE
2 Male   no_college FALSE
3 Female no_college FALSE
4 Male   no_college TRUE
5 Female no_college FALSE
6 Male   no_college FALSE
7 Female college    FALSE
8 Male   college    TRUE
# i 7,763 more rows
```

# 1) Estimate in subgroups

```r
ybar_in_subgroups <- d %>%
  # Group by confounders and treatment
  group_by(sex, a) %>%
  # Summarize mean outcomes and nber of cases
  summarize(ybar = mean(y),
            n = n(),
            .groups = "drop") %>%
  print()
```

```
# A tibble: 4 x 4
  sex    a           ybar     n
  <chr>  <chr>      <dbl> <int>
1 Female college    0.467   896
2 Female no_college 0.102  2953
3 Male   college    0.614   637
4 Male   no_college 0.174  3285
```

# 1) Estimate in subgroups

```
# A tibble: 4 x 4
  sex    a           ybar     n
  <chr>  <chr>      <dbl> <int>
1 Female college    0.467   896
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# 1) Estimate in subgroups

```
# A tibble: 4 x 4
  sex    a           ybar     n
  <chr>  <chr>      <dbl> <int>
1 Female college    0.467   896
2 Female no_college 0.102  2953
3 Male   college    0.614   637
4 Male   no_college 0.174  3285
```

```r
pivoted <- ybar_in_subgroups %>%
  pivot_wider(names_from = a,
              values_from = c("ybar","n")) %>%
  print()
```

```
# A tibble: 2 x 5
  sex    ybar_college ybar_no_college n_college n_no_college
  <chr>         <dbl>           <dbl>     <int>        <int>
1 Female        0.467           0.102       896         2953
2 Male          0.614           0.174       637         3285
```

# 1) Estimate in subgroups

```
# A tibble: 2 x 5
  sex    ybar_college ybar_no_college n_college n_no_college
  <chr>         <dbl>           <dbl>     <int>        <int>
1 Female        0.467           0.102       896         2953
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```

# 1) Estimate in subgroups

```
# A tibble: 2 x 5
  sex    ybar_college ybar_no_college n_college n_no_college
  <chr>         <dbl>           <dbl>     <int>        <int>
1 Female        0.467           0.102       896         2953
2 Male          0.614           0.174       637         3285
```

```r
cate <- pivoted %>%
  mutate(conditional_effect = ybar_college - ybar_no_college,
         n_in_stratum = n_college + n_no_college) %>%
  select(sex, conditional_effect, n_in_stratum) %>%
  print()
```

```
# A tibble: 2 x 3
  sex    conditional_effect n_in_stratum
  <chr>               <dbl>        <int>
1 Female              0.365         3849
2 Male                0.440         3922
```

# 2) Aggregate over subgroups

```
# A tibble: 2 x 3
  sex    conditional_effect n_in_stratum
  <chr>               <dbl>        <int>
1 Female              0.365         3849
2 Male                0.440         3922
```

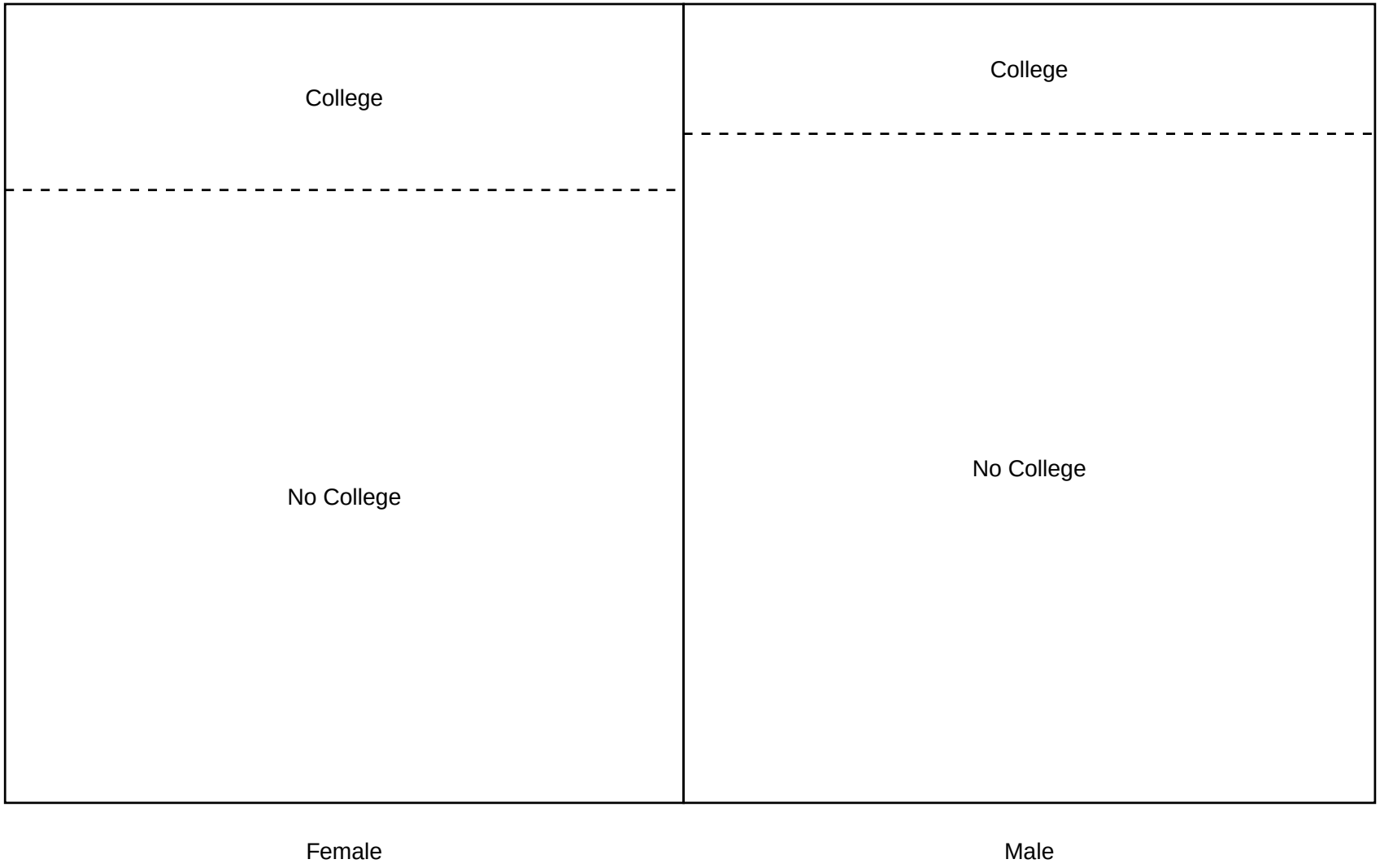
# 2) Aggregate over subgroups

```
# A tibble: 2 x 3
  sex    conditional_effect n_in_stratum
  <chr>               <dbl>        <int>
1 Female              0.365         3849
2 Male                0.440         3922
```

```r
cate %>%
  summarize(population_average_effect = weighted.mean(
    conditional_effect,
    w = n_in_stratum
  ))
```

```
# A tibble: 1 x 1
  population_average_effect
                      <dbl>
1                     0.403
```

# Recap: Intuition

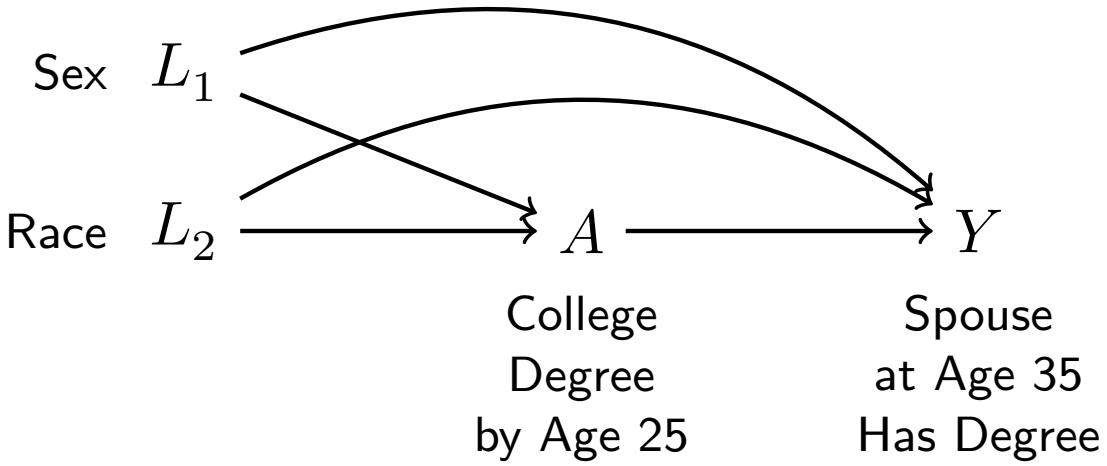


# Recap: In code

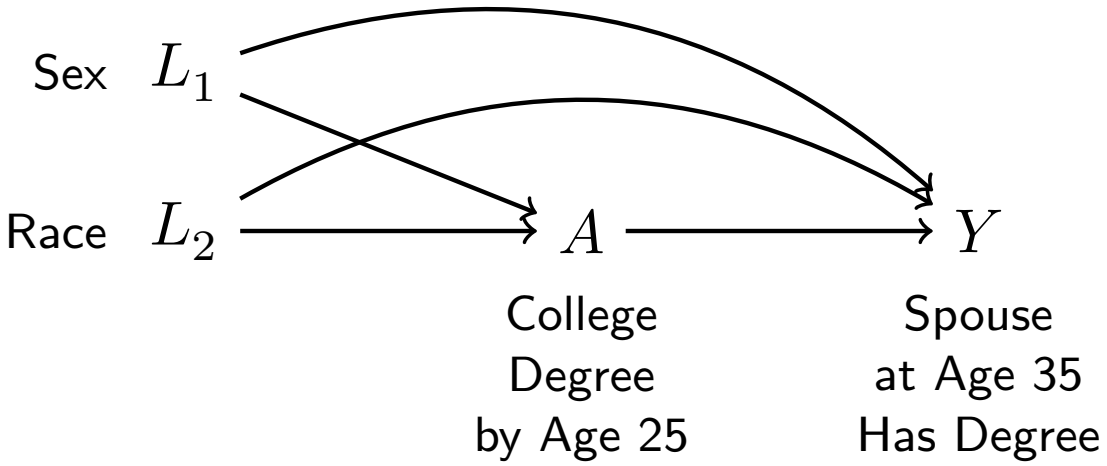
```r
d %>%
  # Group by confounders and treatment
  group_by(sex, a) %>%
  # Estimate within subgroups
  summarize(ybar = mean(y),
            n = n(),
            .groups = "drop") %>%
  pivot_wider(names_from = a,
              values_from = c("ybar","n")) %>%
  mutate(conditional_effect = ybar_college - ybar_no_college,
         n_in_stratum = n_college + n_no_college) %>%
  # Aggregate over subgroups
  summarize(population_average_effect = weighted.mean(
    conditional_effect,
    w = n_in_stratum
  ))
```

```
# A tibble: 1 x 1
  population_average_effect
                      <dbl>
1                     0.403
```

# Adjust for sex and race

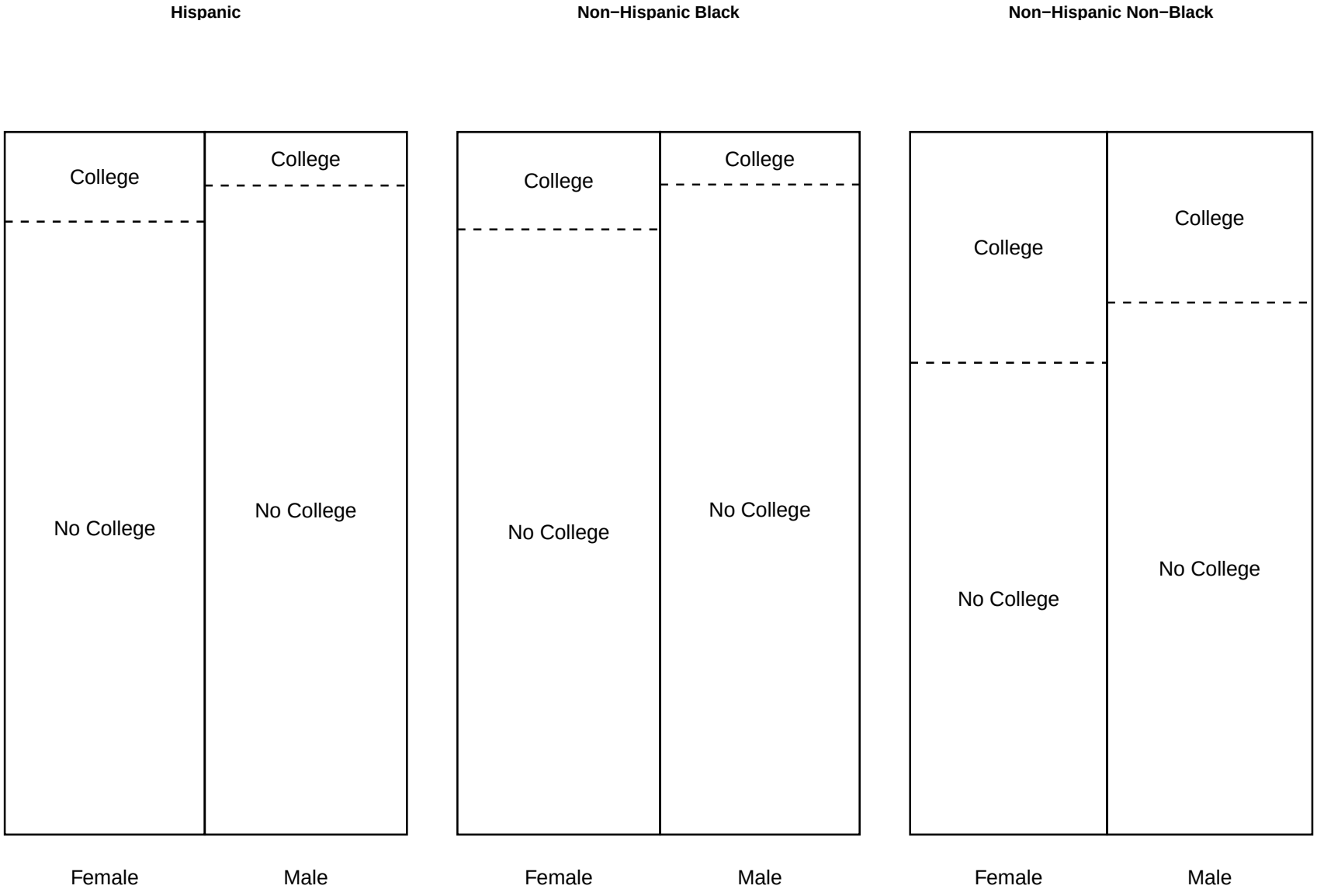


# Adjust for sex and race




1) Estimate effects within subgroups defined by {sex, race}
2) Aggregate over subgroups

# Adjust for sex and race



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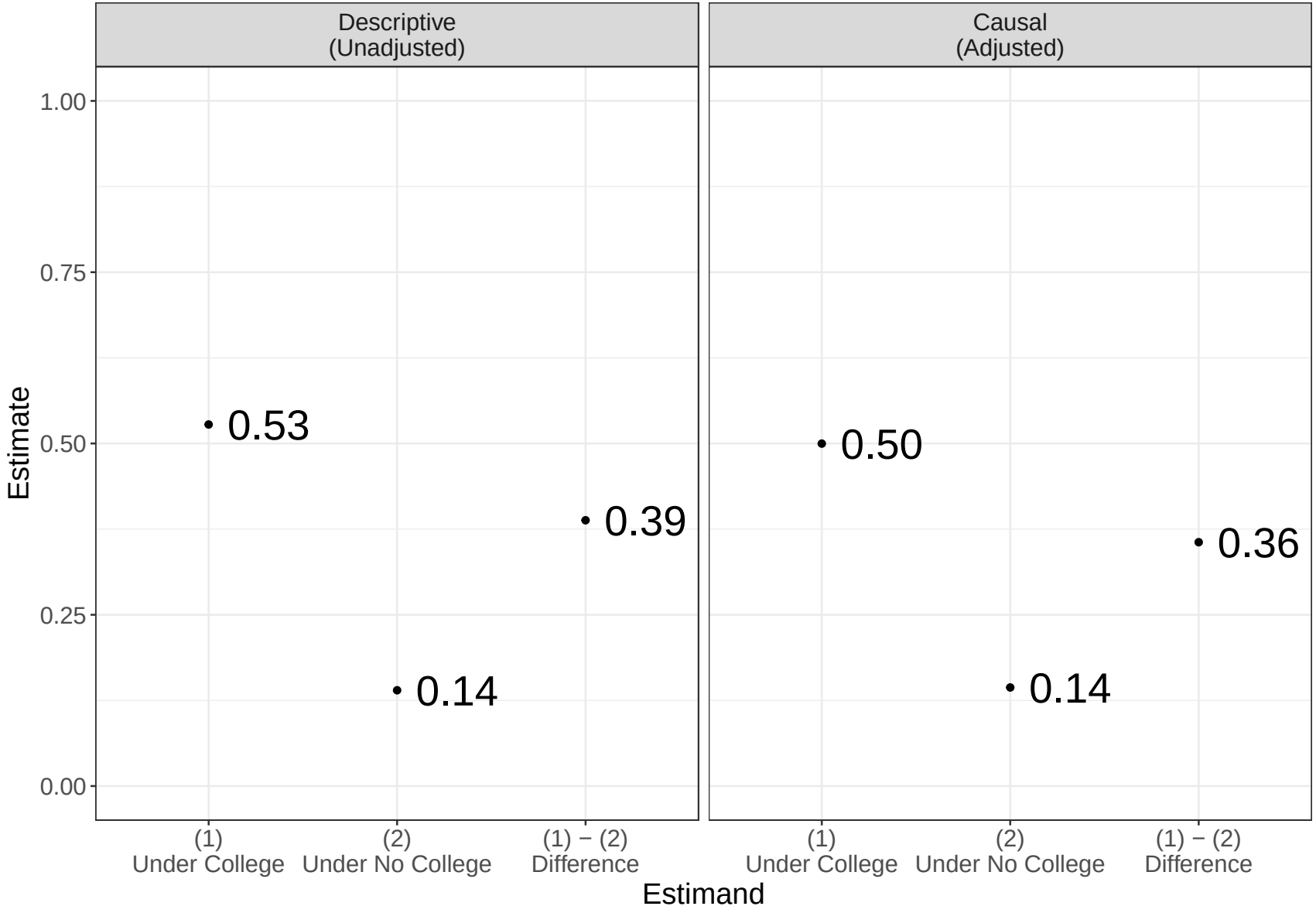
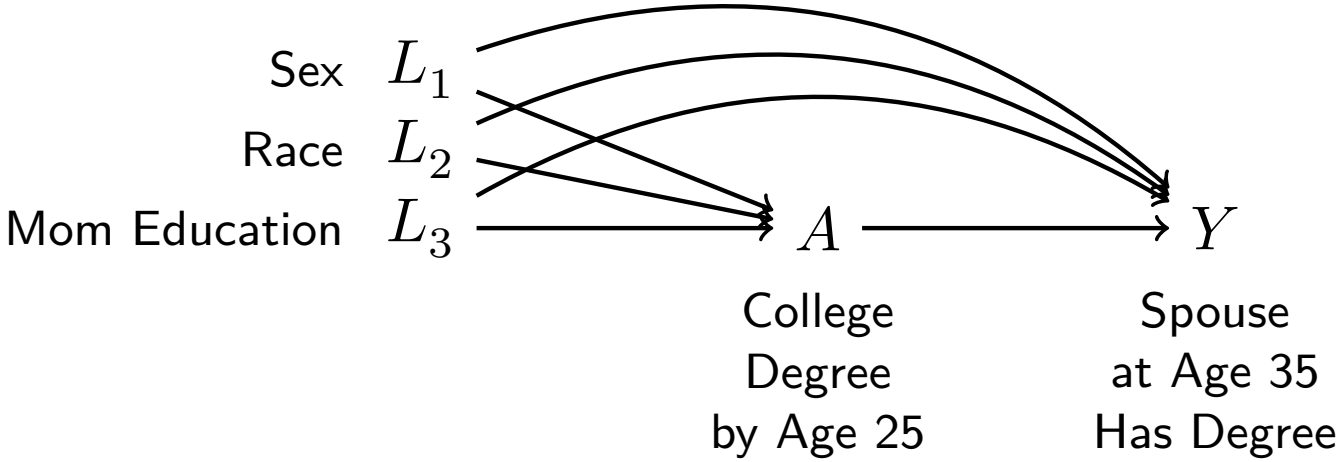
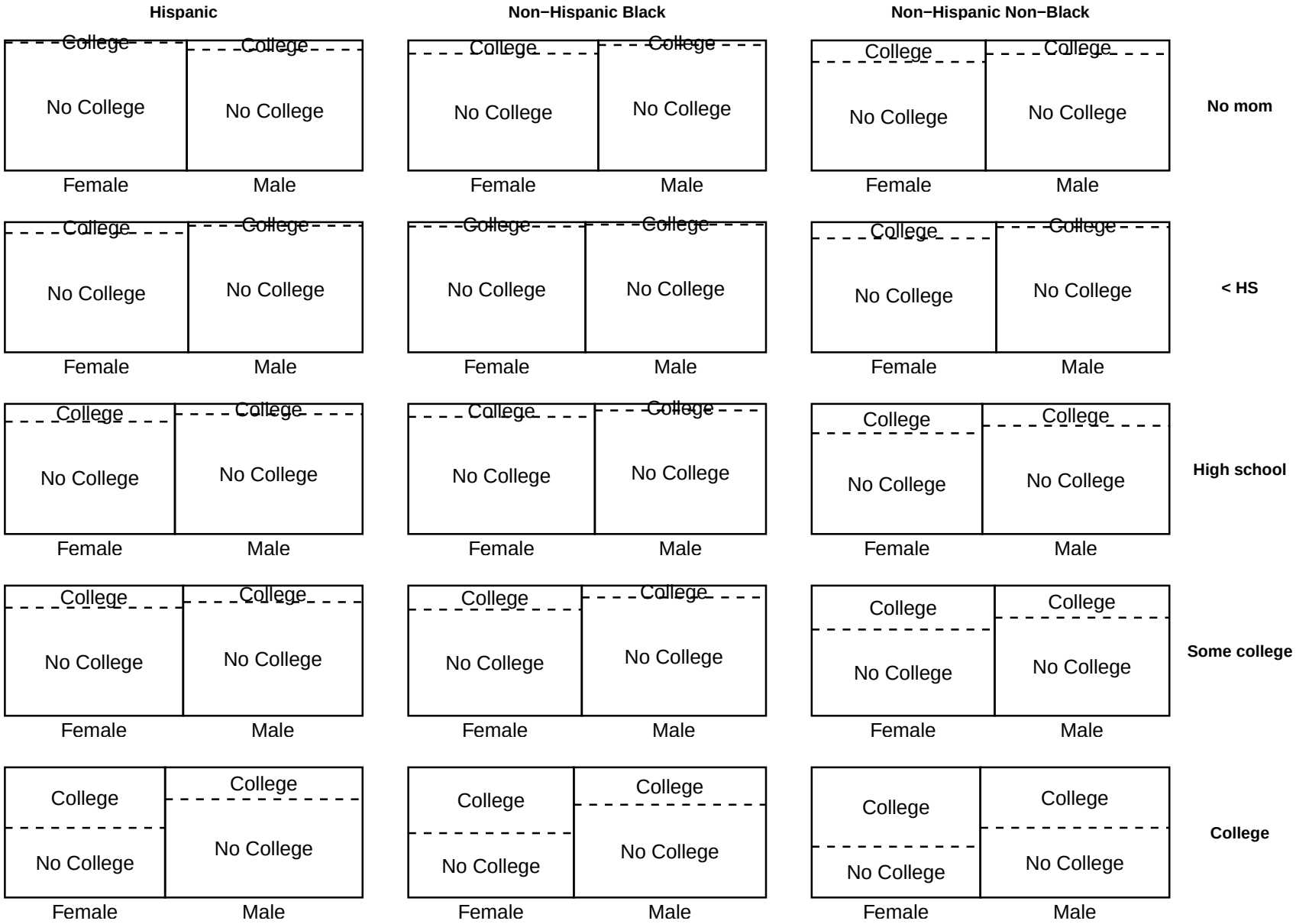


# Adjust for sex, race, mom education

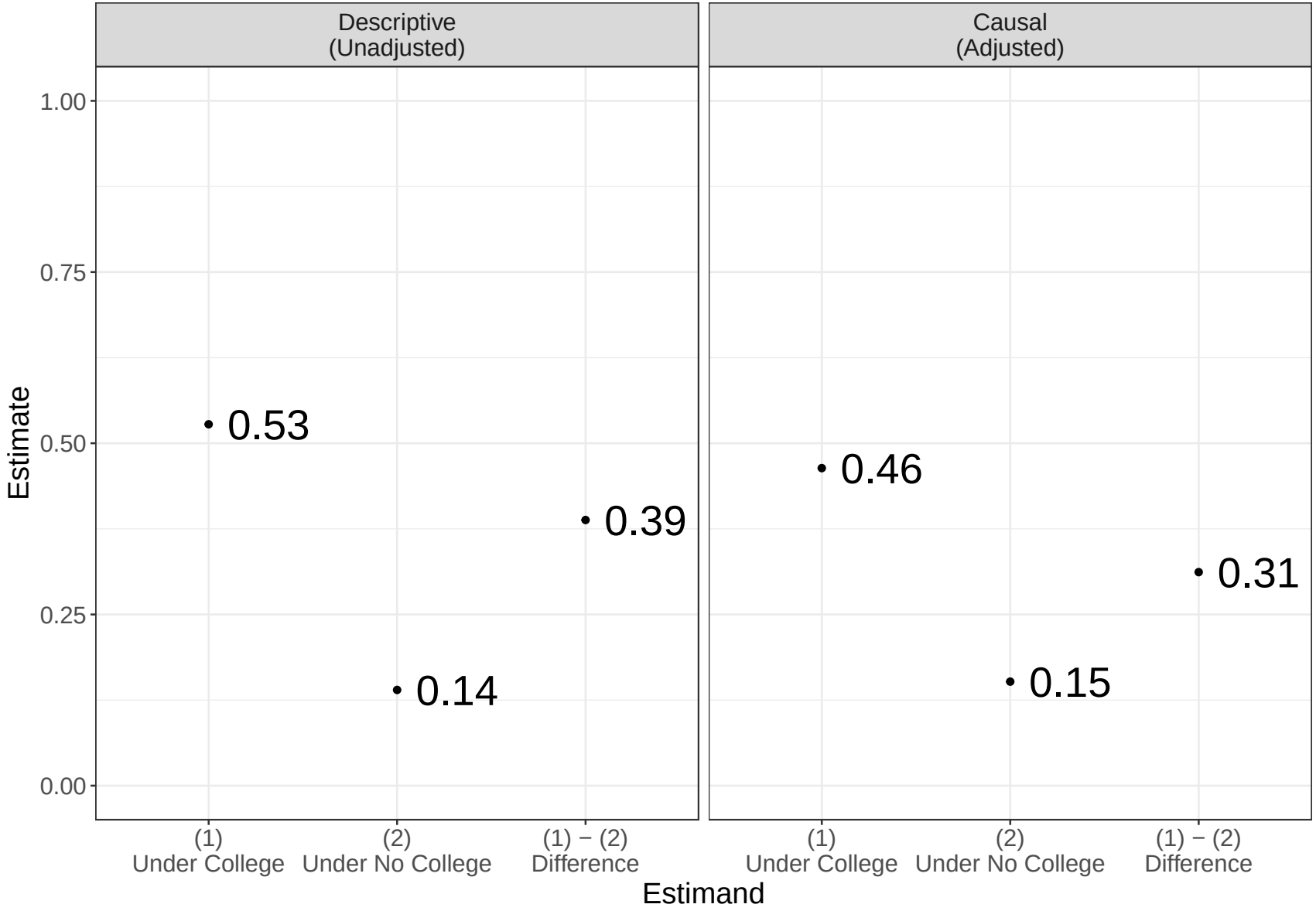



1) Estimate effects within subgroups defined by {race,sex, mom education}
2) Aggregate over subgroups

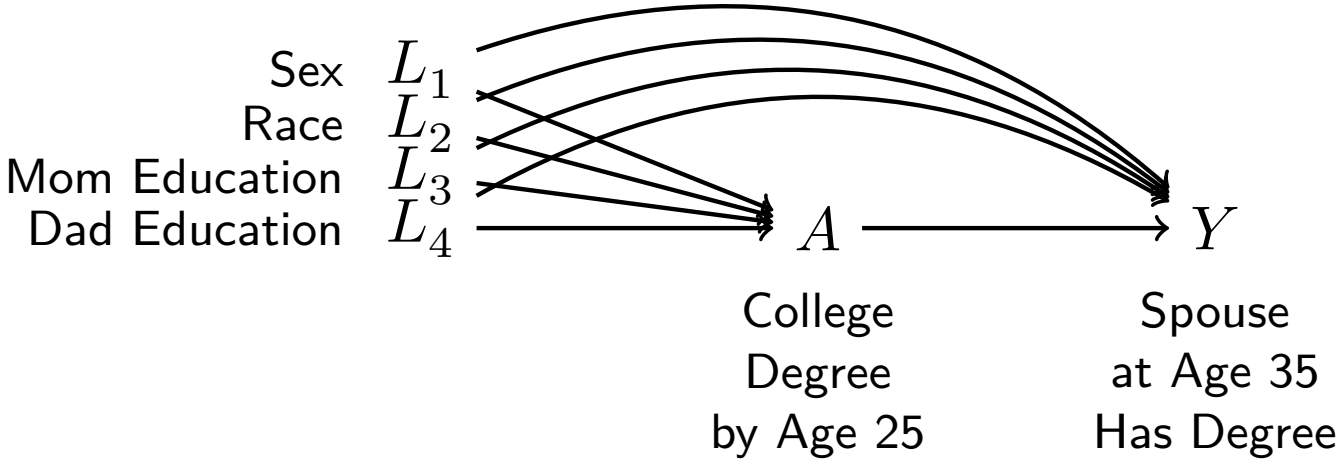
# Adjust for sex, race, mom education



# Adjust for sex, race, mom education

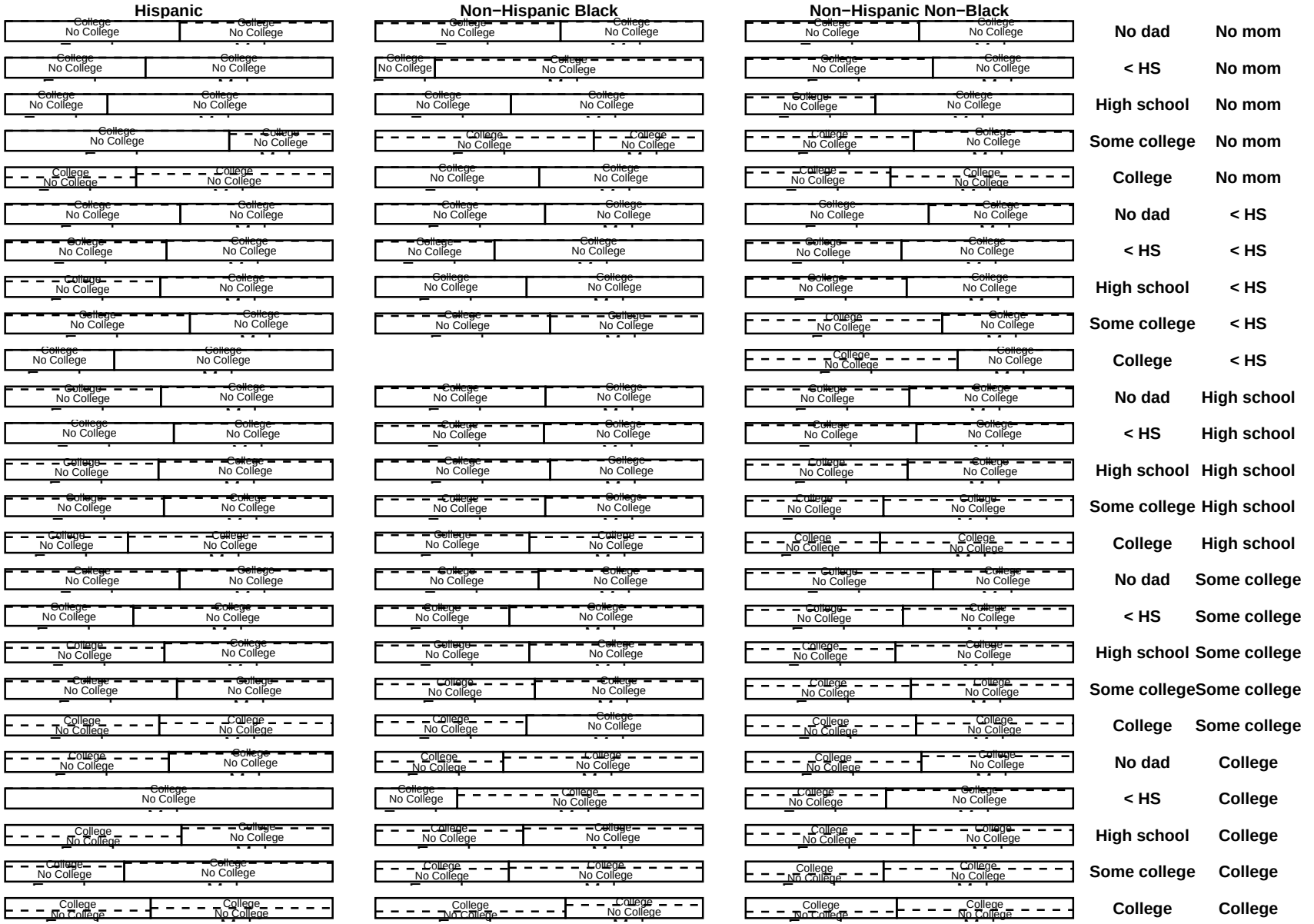


# Adjust for sex, race, mom education, dad education

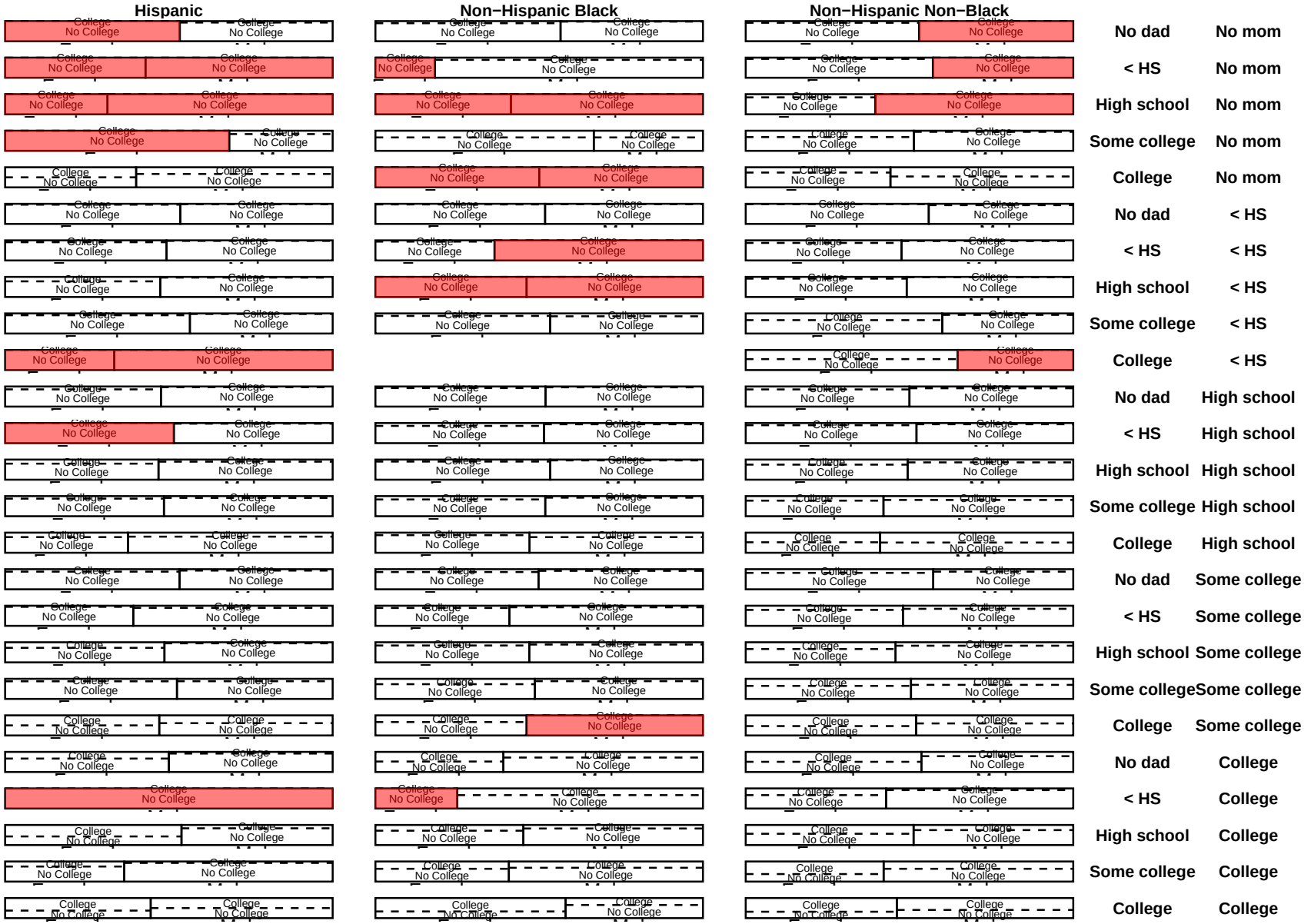



1) Estimate effects within subgroups defined by {race,sex, mom education, dad education}
2) Aggregate over subgroups

# Adjust for sex, race, mom education, dad education



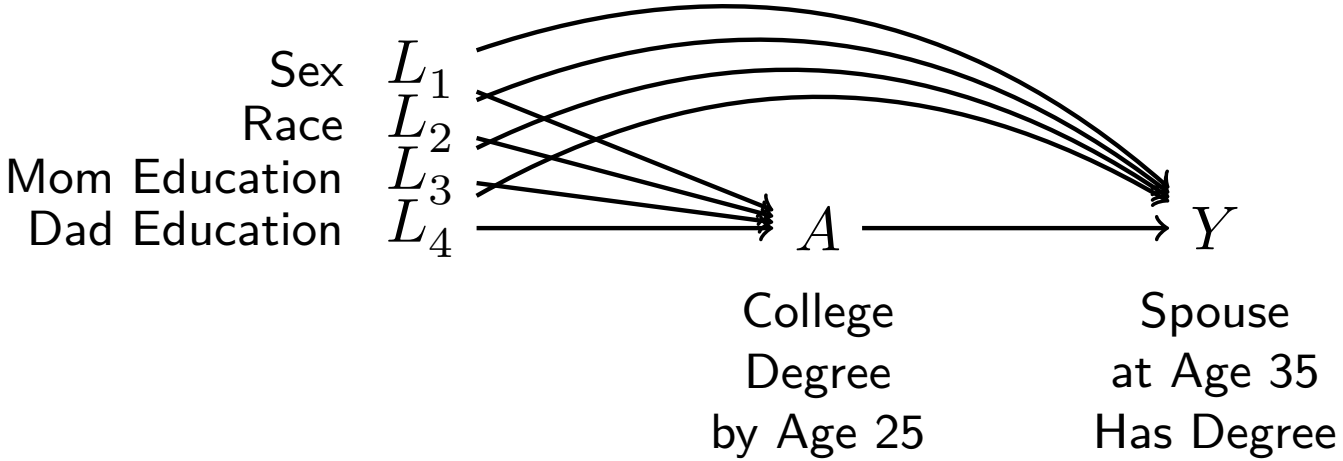
# Adjust for sex, race, mom education, dad education



# Curse of dimensionality: Unpopulated cells

```
# A tibble: 147 x 6
   sex    race    mom_educ    dad_educ     n_college n_no_college
   <chr>  <chr>   <fct>       <fct>            <int>        <int>
 1 Female H       No mom      No dad              NA           32
 2 Female H       No mom      < HS                NA            6
 3 Female H       No mom      High school         NA            5
 4 Female H       No mom      Some college        NA           13
 5 Female H       < HS        College             NA            1
 6 Female H       High school < HS                NA           34
 7 Female Non-H B No mom      < HS                NA            2
 8 Female Non-H B No mom      High school         NA           12
 9 Female Non-H B No mom      College             NA            4
10 Female Non-H B < HS        High school         NA           24
# i 137 more rows
```

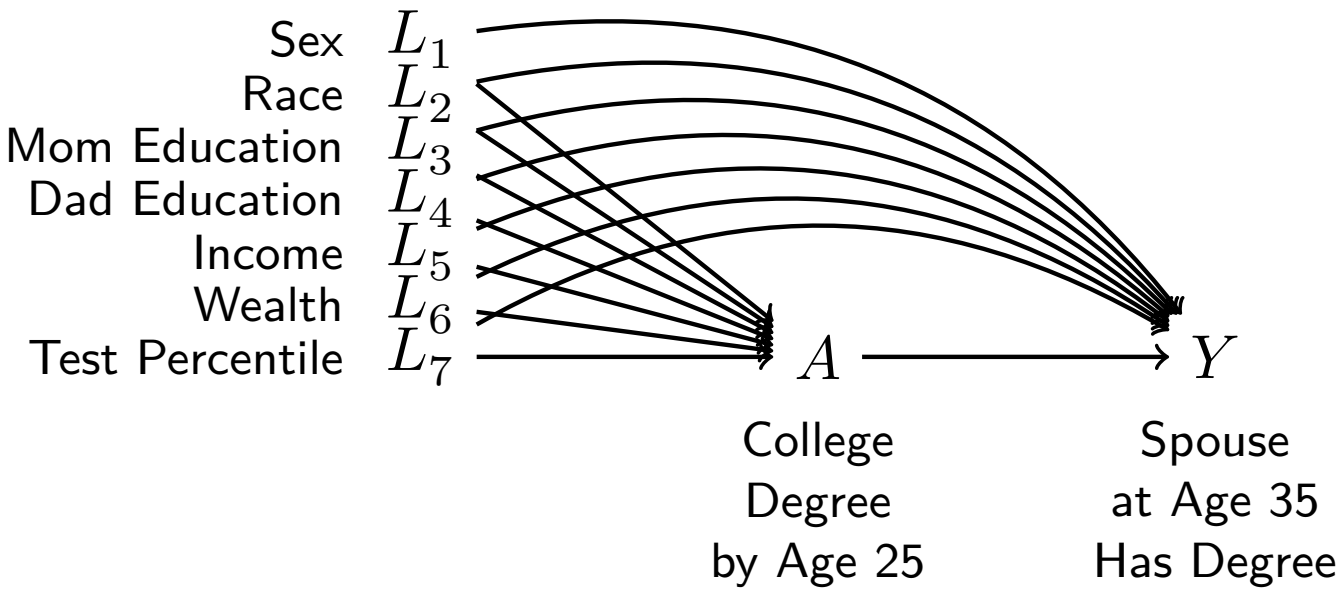
# Curse of dimensionality



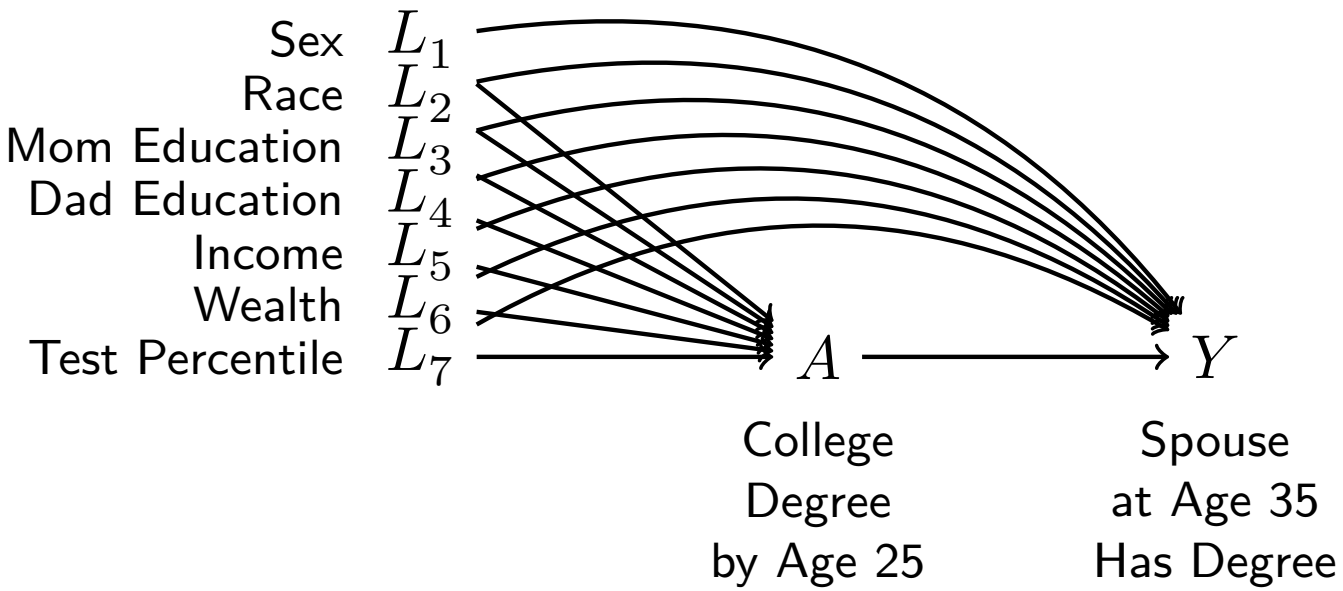

**4.2%** of the sample

is in a subgroup with either 0 treated or 0 untreated units

# Curse of dimensionality



# Curse of dimensionality




**100%** of the sample

is in a subgroup with either 0 treated or 0 untreated units

# Learning goals for today

At the end of class, you will be able to

- explain the curse of dimensionality
- recognize the possible futility of nonparametric estimation

Optionally, read Hernán & Robins Ch 11